

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANGA**

**Master of Computer Application (MCA)
Semester - I University Examination 2009-10
MA701 Foundation of Mathematics**

Date: 31.12.2009, Thursday, Time: 01:00p.m to 04:00p.m Maximum Marks: 70

Notes:-

- (1) This question paper contains two sections **SECTION-I** and **SECTION-II**.
- (2) **SECTION-I** carries **35** and **SECTION-II** carries **35** marks
- (3) All the questions in both sections are compulsory
- (4) Figures to the right indicate the distribution of full marks for the respective question.

SECTION-I

1. Attempt the following:

[5+3+3]

(a) Let $f(x) = (x-2)(8-x)$, $2 \leq x \leq 8$

- (i) What is domain of definition of $f(x)$?
- (ii) Find $f(1-2t)$ and give the domain of definition.
- (iii) Find $f(f(3))$ and $f(f(5))$.
- (iv) Draw the graph of $f(x)$.

(b) Find the area of the region bounded by $y = x$, X-axis and $x=2$, $x=3$.

(c) State and prove De-Morgan's law.

2. Attempt the following :

[4×3=12]

(a) Let $X=\{a, b, c\}$ and $Y=\{1, 2\}$, compute the followings

- (i) $X \times Y$ (ii) $Y \times X$ (iii) $Y \times Y$ (iv) $X \times X$

(b) (i) Define One One function. Also show that the function $f: Q \rightarrow Q$

defined by $f(x) = 7x + 2$ is one one function.

(ii) If Z_1 and Z_2 are two complex numbers, prove that $|Z_1 Z_2| = |Z_1| |Z_2|$.

(c)
$$f(x) = \begin{cases} 3ax+b, & x > 1 \\ 11, & x = 1 \\ 5ax-2b, & x < 1 \end{cases}$$

If f is continuous at $x=1$ then find a and b .

P.T.O.

OR

2. Attempt the following :

[4×3=12]

(a) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 4, 6, 7\}$, find the relation from A to B defined by *greater than equal to*. Also find domain, range and inverse of that relation.

(b) (i) Using definition of limit prove $\lim_{x \rightarrow 4} 7x - 5 = 23$.

(ii) Using Sandwich theorem show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

(c) Evaluate: (i) $(1 + \sqrt{3}i)^{1/2}$ (ii) $(1-i)^{10}$.

3. Attempt the following :

[4×3=12]

(a) (i) If $f(x)$ is differentiable at $x=a$, then show that $f(x)$ is continuous at $x=a$.

(ii) Using definition prove $\frac{d}{dx} 2^x = 2^x \log_e 2$

(b) Evaluate: $\int x^2 \sqrt{a^6 - x^6} dx, (a > 0)$.

(c) Find (i) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^5 - 32}$ (ii) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

OR

3. Attempt the following :

[4×3=12]

(a) Show that the function $f(x) = 1 - \sqrt{1-x^2}$ is continuous on the interval $[-1, 1]$.

(b) Evaluate: (i) $\frac{d}{dx} (x^3 + 5^x + \sin^2 x + \log(2x))$ (ii) $\int (x^2 + 5^x - \frac{1}{x} + \sec^2 x) dx$.

(c) Evaluate: (i) $\int x^2 e^{3x} dx$ (ii) $\int \cos x dx = \sin x + c, \forall x \in R$.

SECTION-II

4. Attempt the following:

[5+3+3]

(a) Solve the following system of equations using matrix inversion

$$x - 2y + 3z = 4$$

$$2x + y - 3z = 5$$

$$-x + y + 2z = 3.$$

P.T.O.

(b) Discuss the probability distribution of binomial with $p=0.75$ and $n=4$.

(c) Calculate the mode for the following data:

Class interval	0-10	10-20	20-30	30-40	40-50
frequency	10	14	19	17	13

5. Attempt the following :

[4×3=12]

(a) Find the rank of $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$.

(b) Solve the equation $\begin{vmatrix} x & 5 & 3 \\ 5 & x+1 & 1 \\ -3 & -4 & x-2 \end{vmatrix} = 0$

(c) Solve: $\left(x - \frac{x}{x+1}\right)^2 + 2x \cdot \frac{x}{x+1} = 3$.

OR

5. Attempt the following :

[4×3=12]

(a) Find the Eigen Values and Eigen Vectors for $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$.

(b) Using properties prove that $\begin{vmatrix} x^2 & y^2 & z^2 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = -(x-y)(y-z)(z-x)$.

(c) If $A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$, show that $A \times B \neq B \times A$.

6. Attempt the following :

[4×3=12]

(a) (i) Find the centre and radius of the sphere $x^2 + y^2 + z^2 - 6x + 8y - 10z + 1 = 0$.

(ii) Find the circle with centre $C(3, -4)$, $r=5$.

P.T.O.

(b) Find the mean deviation from the median for the following:

Marks	0-10	10-20	20-30	30-40	40-50
frequency	8	15	22	15	8

(c) (i) An urn contains four red balls, three green balls and five white balls. If a single ball is drawn, what is the probability that it is green?

(ii) An urn contains 5 red balls and 10 black balls. Eight of them are drawn and placed in another urn. What is the probability that later contains 2 red and 6 black balls?

OR

6. Attempt the following :

[4×3=12]

(a) Find the eccentricity, foci, directrices, length of latus rectum of the

ellipse $\frac{x^2}{49} + \frac{y^2}{36} = 1$.

(b) Calculate mean and standard deviation for the following table giving the age distribution of 542 members

Age in years	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of students	3	61	132	153	140	51	2

(c) In a glass manufacturing process it is known that on an average 1 in every 1000 of items produced has one or more bubbles. What is the probability that a random sample of 5000 will yield less than 2 items possessing bubbles?